

■ Project Task: Sales Data Analysis & BI Dashboard

■ Part 1: Python Analysis & Data Cleaning

Requirements:

- Use Pandas, NumPy, Matplotlib, Seaborn, and Plotly.
- Perform data cleaning: handle missing values, remove duplicates, correct data types, and detect outliers.
- After cleaning, generate 10 insights supported with visualizations (Matplotlib / Seaborn / Plotly).
- Example insights: Best-selling categories, Average order value by payment method, Monthly sales trend, Effect of discounts on sales.
- Export the cleaned dataset to Excel or CSV.

■ Part 2: SQL Analysis

Requirements:

- Import the cleaned dataset into a SQL Database.
- Write SQL queries to answer the following 10 questions:
 - 1 Show the total sales (grand_total) for each year, ordered from newest to oldest.
 - 2 Find the average discount_amount for each payment_method, only for methods with more than 100 transactions.
 - 3 List the top 5 categories by total quantity ordered.
 - 4 Retrieve all orders where the grand_total is greater than the average grand_total of all orders.
 - 5 For each customer, find their first order date (MIN(created_at)) and latest order date (MAX(created_at)).
 - 6 Use a window function to assign a RANK() based on grand_total within each category_name_1.
 - 7 Show the monthly total sales for 2024 only, grouped by M-Y.
 - 8 List the increment_id and grand_total of orders with a discount greater than the average discount.
 - 9 Show customers who joined before 2023 (Customer Since < '2023-01-01') and made more than 5 orders.
 - 10 Use DENSE_RANK() to rank payment methods by their total sales amount.

■ Part 3: Power BI Dashboard

- Connect Power BI to the SQL Database containing the cleaned data.
- Build an Interactive Dashboard that includes: KPI Card (Total Sales), Line Chart (Monthly sales trend), Bar Chart (Sales by category), Pie/Donut Chart (Sales by payment method), Table/Matrix (Top 10 customers by sales), Scatter/Bar (Effect of discount on sales).
- Add filters/slicers for Year, Month, Category, and Payment Method.
- Deliver a Power BI Report (.pbix) with a clean and professional design.

■ Final Deliverables

- Python Jupyter Notebook (data cleaning + 10 insights + charts).
- Cleaned Dataset (Excel/CSV).
- SQL File with 10 queries and their results.
- Power BI Dashboard Report (.pbix).